VoxPreference: a design review of the automated Preference Grammar tool

Demilade Bickersteth · Covenant University · 20 October 2026

Prof. C.U.C. Ugorji's Preference Grammar picks a preferred pronunciation from the forms speakers actually use, by ranking them on six tenets [1, 3]. Done by hand it takes a great many hours, so the team set out to automate it: our ICAEISD 2024 paper and slides [3], Prof's framework for the automation [P2], the PRD [P1], and the user flow in Figure 1 [P3]. My final-year project was one block of that plan, the speech-to-text conversion [8].

This report reviews the whole planned tool, component by component: what the plan said, what made sense, what did not work and why, and what we found and built instead. Each finding points to a published source [n] or to a test in the repository that anyone can rerun (En); both lists are at the end. The tool now runs end to end, from recordings to the operability table, is live on the web, and is reliable today for voiceless and voiced <th> and the KIT vowel (E4).

1. Preference Grammar, in Ugorji's terms

Ugorji proposed PG for codifying the phonological corpora of New Englishes [1, 2]. It has three components [3]. The input is every attested form of a sound, in its words, as speakers produce it, with no underlying forms. The parameters are the six tenets: International acceptance, Contrastiveness, Frequency, Disambiguity, Phonetic simplicity and Pedagogic convenience, ranked by default as I-Ac >> ContR >> FreQ >> DisaM >> Pho-S >> Ped-C. The ranking is strict but violable, as in Optimality Theory [5], and a user may promote, demote or delete a parameter [3]. The output is the preferred forms, the candidates that earn the fewest violations, and the working is shown in an operability table. The published evaluations are *think*, where [θɪŋk] beats [tɪŋk], [sɪŋk] and [zɪŋk] [1: 58; 3], and the GOAT vowel, where [o] wins [4]. One outcome of PG is a Nigerian national model of the vowels [3].

So the automated tool has one job: take recordings to attested forms, attested forms to marks on six ranked tenets, and marks to the operability table and the preferred form. Any automation has to reproduce Ugorji's two worked evaluations.

2. What we found, in short

- **Speech to text, then text to IPA, gives dictionary pronunciations.** That answers "what was said", while PG asks "how was it said" [3, 8]. Because the output always looked plausible, it also hid the training problems (section 4, component 4).
- **Clustering was asked to discover pronunciations** from features that also carry the speaker and the microphone [23].
- **The tenets had definitions but no evidence rules.** The published marks are the analyst's judgement [4], so nothing could be ranked automatically until each tenet had a rule (E3).
- **Success was measured by word error rate.** For PG what matters is how well the variant shares are estimated, checked against human labels with kappa [15] (E4).
- **The infrastructure was planned before the method had been shown to work** [P1].

Hear the sound directly, give every tenet its evidence, and measure against human ears.

3. The planned tool

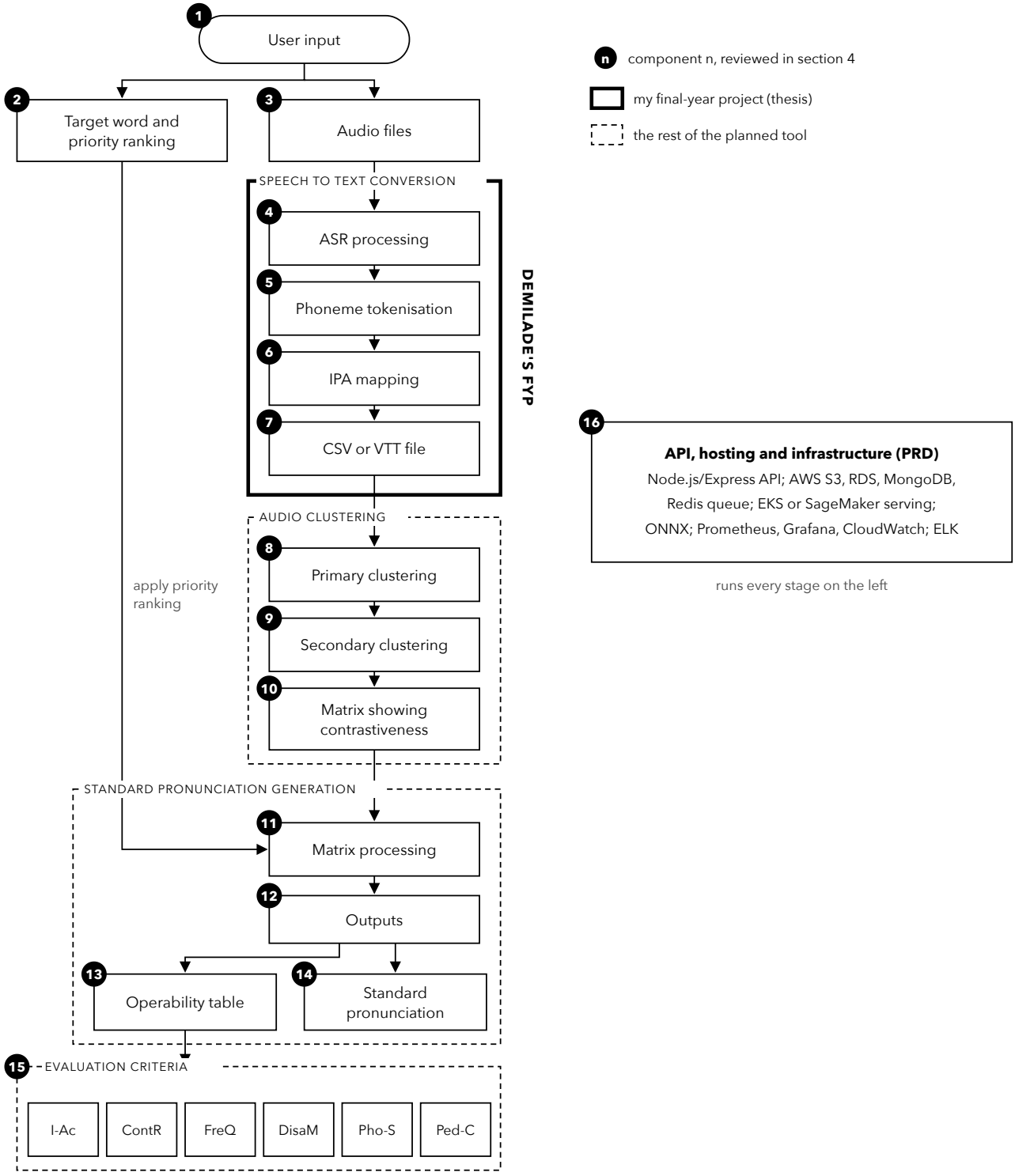


Figure 1. The planned automated PG tool: the team's user flow for the Preference Grammar system [P3], redrawn, with the PRD's infrastructure beside it [P1]. The numbers match the components in section 4. The block in the thick outline, speech-to-text conversion, was my final-year project [8].

The aim was right. Most of what went wrong sat between the boxes: what each one had to hand to the next.

4. Component by component

Input

1 User input

THE PLAN	A web system where a user uploads speech and sets a tenet preference [P2]; the PRD adds user stories for researchers, educators, developers, policy makers and learners [P1].
MADE SENSE	Putting the user's ranking at the centre. PG lets a user promote, demote or delete parameters [3], so the tool has to let them.
WHAT DID NOT WORK, AND WHY	Five audiences and their features were planned before a single analysis had run end to end [P1]. That spread the effort thin at the start.
WHAT WE FOUND, AND WHAT THE TOOL DOES NOW	One page for one job: words, recordings and a ranking in; the operability table and the preferred form out. The other audiences can build on it once it is trusted.

2 Target word and priority ranking

THE PLAN	The user gives a target word and a priority ranking; the ranking is applied at matrix processing [P3].
MADE SENSE	PG's input is attested forms within their lexical contexts [3], and the word is that context. The ranking belongs to the user.
WHAT DID NOT WORK, AND WHY	One word yields few tokens. The plan did not say which sound in the word is under study, and a ranking kept as a plain list cannot pool two tenets or delete one, which PG allows [3].
WHAT WE FOUND, AND WHAT THE TOOL DOES NOW	Several words that share a sound; the sound is chosen from a list (or inferred) and each one is marked validated or not. The ranking editor reorders, deletes and pools tenets; pooled and re-ranked evaluations are tested (E3).

3 Audio files

THE PLAN	Accept large continuous speech files [3]. Prof's framework reviews continuous speech separation for overlapping talk [P2].
MADE SENSE	Real speech from many speakers, including long recordings, is what PG needs.
WHAT DID NOT WORK, AND WHY	Each file was treated as one voice: the original service returned one transcription per file (E2), and speaker separation was reviewed [P2] but not yet designed. A podcast panel would have counted as one speaker.
WHAT WE FOUND, AND WHAT THE TOOL DOES NOW	Any common audio format, voice notes recorded in the browser, and podcasts. Speaker separation with ECAPA-TDNN [12, 13] credits each word to the person who said it; on 8 held-out conversations it found the right number of speakers every time (E6).

Speech to text conversion: my final-year project

4 ASR processing

THE PLAN	PRD: a CNN-RNN trained on Colab with MFCC features [P1]. My thesis: wav2vec 2.0 [9] fine-tuned on Nigerian English, SLR70 (5.77 hours, 3,359 utterances, 31 speakers) [8, 18].
MADE SENSE	Fine-tuning a pretrained wav2vec 2.0 model on public Nigerian recordings was a sound, current choice, and better than training a CNN-RNN from scratch on about 6 hours of audio. A random 90/10 split and word error rate are the usual practice in speech-recognition work, and the thesis measured exactly what it set out to measure [8].

WHAT DID NOT WORK, AND WHY	In hindsight, three things stopped the fine-tune from learning, none of them visible in the output. (a) The script lowercased the transcripts (E2), but the model's vocabulary has only capital letters, so all 174,915 label characters in the training half became `<unk>` and none was a real letter (E1). (b) The installed transformers 4.51.3 leaves one weight, `masked_spec_embed`, uninitialised when the model loads (it prints a warning to that effect) (E1). The published training logs show what followed: at all 37 logged steps the loss reads 0.0, the gradient norm is NaN and the learning rate is 0, over 3,780 steps (E1). Compared tensor by tensor, all 212 weights the published model shares with the stock model are identical, and its one extra weight, `masked_spec_embed`, holds 41 NaN values out of 768 (E1). (c) The split put all 31 speakers on both sides, and 67 of the 336 validation sentences also occur in training (E1). Re-run on that same 336-utterance validation split, the unchanged stock model scores WER 36.7% with 112 deletions (E1); the thesis reports 41.10% with 112 deletions [8]. With every weight identical, the reported figure describes the stock model; the difference in WER most likely comes from how the transcripts were normalised before scoring. Underneath all three, speech recognition answers which words were said, while PG asks how they were said.
WHAT WE FOUND, AND WHAT THE TOOL DOES NOW	Speech recognition now only finds the words: Whisper [11] searches unscripted audio, including look-alike words (*tree* for *three*), and a letter aligner places the words when a transcript is given. The fine-tuning script is fixed (capital-letter labels, the weight re-initialised, a guard against invalid losses, a speaker-disjoint split) and a 30-step check ran cleanly (loss 1.40 to 0.72). It has not been rerun in full because the tool no longer depends on it; a full run would take 2 to 3 hours per pass on this laptop, or about an hour on a free Colab GPU.

5 Phoneme tokenisation

THE PLAN	Split speech into phoneme tokens before clustering [P2, P3].
MADE SENSE	PG works on segments inside words, so the tool must find the segment.
WHAT DID NOT WORK, AND WHY	In the thesis the tokens came from text: the transcription was split into words (thesis Figure 3.1) [8]. Nothing in the audio was tokenised.
WHAT WE FOUND, AND WHAT THE TOOL DOES NOW	The pronouncing dictionary [21, 22] says where the sound sits in each word, and the phone model's alignment finds its frames in the audio.

6 IPA mapping

THE PLAN	Map words or phonemes to IPA: eSpeak in the thesis [8], a dictionary such as CMU in the team's stage plan [P4].
MADE SENSE	IPA is the shared notation, and PG tables are written in it. eSpeak is the standard way to turn text into IPA, and the thesis itself presents its IPA output as illustrative, not verified (thesis §4.4) [8].
WHAT DID NOT WORK, AND WHY	A dictionary gives the citation form. The thesis's Table 4.2 gives *there* as [ðɛr] [8], and the same entry is returned whether the speaker said [ð] or [d]. PG's input is the attested forms [3], so this step removes the variation PG ranks.
WHAT WE FOUND, AND WHAT THE TOOL DOES NOW	Candidate pronunciations, written in IPA, are scored against the audio by a multilingual phone model [10], the goodness-of-pronunciation approach [14], with a confidence for each token. Each sound carries a validation status from phonetician-labelled speech (section 6) (E4).

7 CSV or VTT file

THE PLAN	Write text and IPA to a CSV, a 5×6 matrix in the framework [P2], or a VTT subtitle file [P3].

MADE SENSE	Plain files an analyst can open.
WHAT DID NOT WORK, AND WHY	A VTT file needs word times. The thesis estimated them from the clip's duration (thesis Figure 3.1) [8]; the deployed service returned no word times at all (E2). The 5×6 size came from the matrix, not from the data.
WHAT WE FOUND, AND WHAT THE TOOL DOES NOW	Downloads in CSV, JSON, HTML and subtitle formats, with measured start and end times for every token (E8).

Audio clustering

8 Primary clustering

THE PLAN	K-means (and spectral clustering) on MFCC features to group similar sounds [P3, 3].
MADE SENSE	Unsupervised grouping could find variants without labelled data, a sensible instinct.
WHAT DID NOT WORK, AND WHY	MFCC features carry the speaker's voice and the recording channel as well as the sound; speaker-identification systems are built on them [23]. Clusters of whole words can therefore follow the speaker more than the pronunciation, and a cluster does not say which variant it holds.
WHAT WE FOUND, AND WHAT THE TOOL DOES NOW	The phone model's variant label does the grouping: every token is [θ], [t], [s] and so on, with a confidence (E4).

9 Secondary clustering

THE PLAN	Sub-clusters by cosine similarity, at most a 5×6 matrix [3, P4].
MADE SENSE	Looking for finer detail inside a variant.
WHAT DID NOT WORK, AND WHY	The same limits as above, and the matrix size decided the number of clusters.
WHAT WE FOUND, AND WHAT THE TOOL DOES NOW	Acoustic sub-clusters inside each variant, from the phone model's features. In our checks they barely separate the variants, so the app shows them as exploratory (E7).

10 Matrix showing contrastiveness

THE PLAN	A confusion matrix of the sub-clusters, used for the operability analysis [3].
MADE SENSE	Contrast matters to PG: Contrastiveness is one of the six tenets [3].
WHAT DID NOT WORK, AND WHY	A confusion matrix measures how much clusters overlap acoustically. PG defines Contrastiveness as phonological distinctiveness and Disambiguity as avoiding ambiguity [3]: whether a variant merges a contrast and makes words sound alike.
WHAT WE FOUND, AND WHAT THE TOOL DOES NOW	The contrast matrix is kept as exploratory (E7). Contrastiveness is marked from the Lingua Franca Core [6] and Disambiguity from pronouncing dictionaries [21, 22]: [t] in *think* merges 78 common word pairs, such as *bath* and *bat* (E3).

Standard pronunciation generation

11 Matrix processing

THE PLAN	Apply the user's ranking to the matrix [P3]; assign a rank to each cluster by quality or relevance [P4].
MADE SENSE	The ranking is the heart of PG [3].

WHAT DID NOT WORK, AND WHY	PG's published evaluations are the analyst's judgement and give no algorithm for awarding a mark; Jowitt asks on what basis International acceptance is assigned [4]. Without such rules nothing could be ranked automatically, and ranking clusters by quality is a different operation from PG's evaluation [3, 5].
WHAT WE FOUND, AND WHAT THE TOOL DOES NOW	A rule for each tenet, chosen so that Ugorji's *think* and GOAT evaluations [1, 3, 4] come out exactly, under strict domination as in Optimality Theory [5] (E3). A variant competes once it has at least 3 tokens from 2 speakers. Experts can override any mark.

12 Outputs

THE PLAN	The operability table and the standard pronunciation, saved as .txt and .csv; the stage plan adds a "PG operability metric" of accuracy, consistency and usability [P4].
MADE SENSE	Two outputs, exactly as PG defines them [3].
WHAT DID NOT WORK, AND WHY	Accuracy and usability describe the software. In PG, operability is the evaluation of candidates against the ranked tenets [3], so the two are better kept apart.
WHAT WE FOUND, AND WHAT THE TOOL DOES NOW	Both outputs, plus warnings when a sound is not validated or the data is thin, and every token listed for checking by ear (E8).

13 Operability table

THE PLAN	Ugorji's tableau: candidates in rows, tenets in columns, violation marks in the cells [1, 3].
MADE SENSE	It is how PG shows its working.
WHAT DID NOT WORK, AND WHY	Not yet produced by the earlier work.
WHAT WE FOUND, AND WHAT THE TOOL DOES NOW	Produced in PG style, with an OT view. Figure 3 is the *think* table as the tool prints it, mark for mark as in Ugorji's slide [3] (E3).

14 Standard pronunciation

THE PLAN	The winning candidate [P3].
MADE SENSE	PG's output component: the preferred forms earn the fewest violations [3].
WHAT DID NOT WORK, AND WHY	The plan did not say how sure the winner is.
WHAT WE FOUND, AND WHAT THE TOOL DOES NOW	The preferred form, its tier (formal standard, allophone or outer norm), and how often it still wins when speakers are resampled and uncertain tokens redrawn (E8).

Evaluation criteria

15 Evaluation criteria: the six tenets

THE PLAN	International acceptance, Contrastiveness, Frequency, Disambiguity, Phonetic simplicity and Pedagogic convenience, ranked I-Ac >> ContR >> FreQ >> DisaM >> Pho-S >> Ped-C [3].
MADE SENSE	These are Ugorji's own definitions and default ranking [1, 3].
WHAT DID NOT WORK, AND WHY	The definitions are qualitative, and the sources leave open what evidence decides each mark [4]. Frequency in particular can be read two ways, and the reading changes the winner for *think* (E3).

WHAT WE FOUND, AND WHAT THE TOOL DOES NOW	A source of evidence for each tenet (section 7): a knowledge base, the Lingua Franca Core [6], the recordings, pronouncing dictionaries [21, 22] and PHOIBLE [16]. Both readings of Frequency are available as a setting.

Around everything

16 API, hosting and infrastructure

THE PLAN	A Node.js/Express API; AWS S3, RDS (PostgreSQL), MongoDB and a Redis queue; serving on EKS or SageMaker; ONNX; Prometheus, Grafana, CloudWatch and ELK [P1].
MADE SENSE	An API, and containers.
WHAT DID NOT WORK, AND WHY	The full estate was planned before the method had been shown to work, and it is more than the analysis needs at this stage.
WHAT WE FOUND, AND WHAT THE TOOL DOES NOW	One Python service serves the page and the API from one Docker image, live on Hugging Face's free hardware [19] (section 6) (E9). One GitHub repository, deploy on push and a daily keep-awake visit are being set up.

5. The tool now

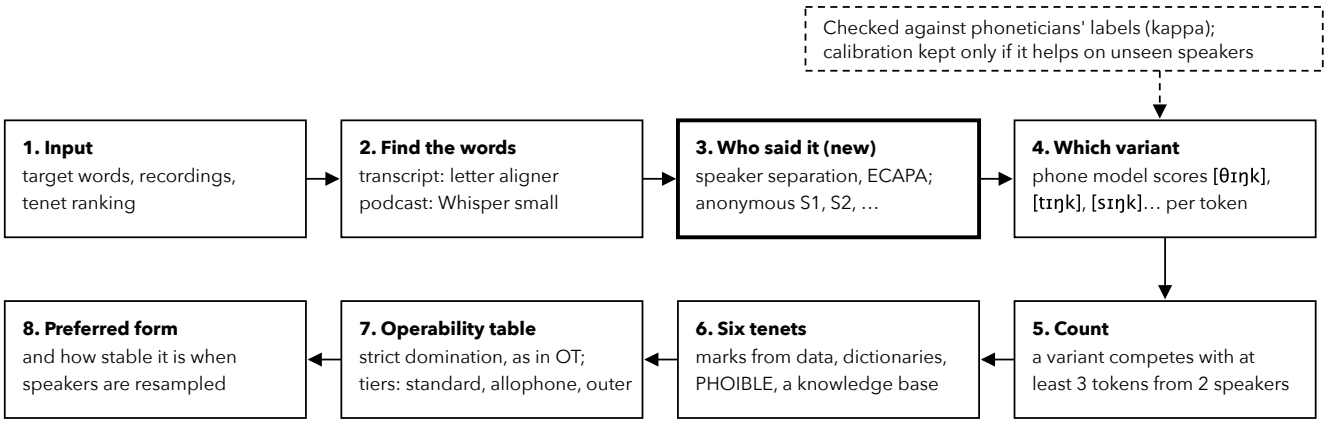


Figure 2. The full cycle as it runs now. Every step is in `backend/pg/`; the web page, the API and the command line run the same code (E8).

The models

Job	Model	Why this one
Hear which variant was said	`facebook/wav2vec2-xlsr-53-espeak-cv-ft`	A phone recogniser trained on many languages, so it can hear [t] against [θ], not just English words. It compares candidate pronunciations of the word against the audio.
Find each word in a sentence	`facebook/wav2vec2-base-960h`	Lines a known transcript up with the audio, letter by letter.
Find words in podcasts	`openai/whisper-small`	Writes a transcript with word times; each hit is then re-aligned. It also searches look-alike words (*tree* for *three*).
Tell the speakers apart	`speechbrain/spkrec-ecapa-voxceleb`	Speaker fingerprints for short stretches of speech, grouped into speakers (section 6).

Sources: phone model [10], built on wav2vec 2.0 [9]; aligner [9]; Whisper [11]; ECAPA-TDNN [12] through SpeechBrain [13]; all loaded with the transformers library [20] or SpeechBrain.

6. Results

How well it hears each sound

The only free Nigerian English recordings with pronunciations transcribed by trained phoneticians are 22 speakers in the Speech Accent Archive [17], all reading the same paragraph: 753 test tokens. The fairest single number is Cohen's kappa [15], agreement beyond chance: a model that always answers the commonest variant scores 0, however good its accuracy looks (E4).

Sound	Tokens	Accuracy	Always guess the commonest	Kappa	Status
<th> /θ/ (think)	85	82%	58%	0.68	Validated
<th> /ð/ (this), calibrated	129	81%	53%	0.64	Validated
KIT (bring, six)	112	81%	76%	0.55	Validated
NURSE (her)	86	43%	41%	0.23	Not validated
GOAT (go, snow)	68	72%	87%	0.16	Not validated
FACE (snake, train)	89	54%	72%	0.13	Not validated
FLEECE (please, meet)	183	90%	92%	0.06	Not validated

An independent review was asked to break this work, not defend it (E7). It found two calibrations that only looked useful (GOAT and FACE gave every token the same answer), a weak test ("same preferred form as the experts"), and a phone model that leans towards dictionary forms: it hears Nigerian [d] as [ð] 36 times and the reverse once. All are fixed or stated plainly; a calibration is now kept only when kappa rises on speakers it was not fitted on, which leaves voiced <th> alone.

What it found in Nigerian speech

On 1,519 sentences from Google's Nigerian English corpus [18], 31 speakers (E5):

Sound	Variant shares	Preferred (Ugorji's ranking)
<th> /θ/, 618 tokens	θ 52%, t 36%, d 5%, s 4%, f 2%. Word-initially θ 48% and stops 48%, close to ICE-Nigeria (θ 47%, t 49%).	[θ], in 100% of resamples
<th> /ð/, 925 tokens	d 57%, ð 40% once the model's bias is corrected	[ð] (99%); *d* loses on Disambiguity, not on the data
NURSE, 676 tokens (not validated)	ɜː 36%, ɛ 22%, ɔ 17%, a 15%, e 8%; the model still leans to the dictionary vowel	[ɜː]

ICE-Nigeria comparison: Oyebola, Ho & Li [7].

Speaker separation

PG counts speakers, not files. The tool cuts speech into 1.5-second windows, takes a speaker fingerprint of each with ECAPA-TDNN [12, 13], removes what all the fingerprints share (room, microphone, language), averages each window with its neighbours, groups them into speakers, and credits every target word to the person speaking at that moment. It was tested on made-up 4-minute panels of 2 to 5 real SLR70 Nigerian speakers [18], including three and four women together (E6). An earlier model, WavLM, credited under 75% of words correctly for four women even when told the number of speakers; ECAPA was then tuned on 4 conversations and tested on 8 it had never seen.

Test	Result
Held out: 8 conversations, 2 to 5 speakers (E6)	right number of speakers in 8 of 8; 100% of sampled words credited to the right speaker
Live server: a 63-second, 3-speaker conversation (E9); a 3-speaker check also runs on every push (E10)	3 speakers found; 5 of 5 target words credited right; 48 seconds

The test is kinder than a real podcast: each clip was recorded separately, so microphone differences help, and the speakers come from the same pool of 31.

Live deployment

The tool is live at `https://thebickersteth-voxpreference.hf.space` on Hugging Face's free CPU hardware (2 cores, 16 GB) [19]. New Docker Spaces have needed a paid plan since mid-2026 [19]; ours existed before that and keeps the free hardware, so it must not be deleted. The account has no payment method and cannot be charged. Twelve SLR70 clips from 6 speakers ran in 18 seconds (6 [θ], 4 [t], [θ] preferred) (E9). A free Space sleeps after 48 hours without visitors and wakes in a minute or two [19].

Podcast test

Results pending. A 43-minute Nigerian panel podcast with four hosts, two men and two women, is being analysed for every sound, once with Ugorji's ranking and once with Frequency first. The results go here when the run finishes.

7. The tenets, and Ugorji's table as the tool prints it

No PG source gives an algorithm; the published marks are the analyst's judgement [4]. The rules below are our proposal, chosen so that both of Ugorji's evaluations [1, 3, 4] come out exactly right (E3). They are defaults the PG experts can change, not PG's own verdicts.

Tenet	A variant gets a mark when	Evidence
International acceptance	it is not an accepted international norm for the sound (for /θ/: θ, t, t̪, f are; s, z are not)	knowledge base: RP and GA, established L1 varieties, the Lingua Franca Core
Contrastiveness	it merges a contrast that matters for international intelligibility	Lingua Franca Core
Frequency	its share is under 20%, or, in "most frequent" mode, it is not the commonest variant	the recordings
Disambiguity	it makes common words sound alike (*three* and *tree*)	pronouncing dictionaries and word frequency
Phonetic simplicity	it is missing from most of Hausa, Igbo and Yoruba, or rare in the world's languages	PHOIBLE
Pedagogic convenience	it is missing from those languages and rarely used by the speakers analysed	PHOIBLE and the recordings

Sources: Lingua Franca Core [6]; pronouncing dictionaries [21, 22]; PHOIBLE [16].

<th> in *think*	Share	I-Ac	ContR	FreQ	DisaM	Pho-S	Ped-C	Tier
a. tɪŋk	48.8%				*!			allophone
b. ☞ θɪŋk	47.2%					*		preferred
c. sɪŋk	4.0%	*!		*	*			outer norm
d. zɪŋk	0.0%	*!		*	*			outer norm

Figure 3. Ugorji's *think* evaluation [1: 58; 3] as the tool produces it, with his default ranking: the same marks and the same winner as his slide (E3). Shares are word-initial /θ/ in ICE-Nigeria [7]. The reasons the tool gives: [t] makes 78 common word pairs sound alike (*bath/bat*) [21, 22], [θ] is absent from Hausa, Igbo and Yoruba [16], and [s] and [z] are outside the international norms [6]. Shaded cells no longer matter once a fatal mark (!) is given [5].

What "rank by frequency" means

Many users read "Frequency first" as "the commonest pronunciation wins": if *think* came out with [θ] 49 times and that was the most, [θ] is preferred. Ugorji's *think* table only comes out right under another reading, where Frequency removes rare variants (under 20%), because in the ICE-Nigeria shares [t] is slightly more common than [θ] [7]. Both readings are in the tool, and the test suite checks that each gives its expected winner (E3). The default follows Ugorji.

8. Questions for Prof. Ugorji

- Which norms count as international, and why [t] escapes that mark in *think* while [s] does not [1, 3]. Jowitt raises the same question [4].
- What Frequency means: a threshold or "the commonest wins"; per token or per speaker; over whose speech. This alone can flip *think* to [t] (E3).
- What separates Contrastiveness from Disambiguity, and what counts as evidence for Pedagogic convenience. The full list of 10 is in `backend/pg/RESEARCH.md`, section 6.

9. Next steps

1. Label 150 to 300 new Nigerian tokens, ideally twice, with the listening tool (`python -m pg.review`). This turns the vowels from not validated to validated.
2. Take the questions above to Prof. Ugorji before presenting automated tables as PG results.
3. Add the podcast results and check the speaker separation against the known four-person panel.
4. Try a stronger model for vowels: the Montreal Forced Aligner's English model, trained partly on Nigerian English, or formant measurements.
5. Finish the single repository with deploy on push and the daily keep-awake visit.

The method works where it has been checked. The next work is checking more of it.

References

1. Ugorji, C.U.C. (2010). *Nigerian English Phonology: A Preference Grammar*. Frankfurt am Main: Peter Lang. (The *think* evaluation is on p. 58.)
2. Ugorji, C.U.C. (2013). Standardising New Englishes: A suggestion for phonological corpora. *Journal of English as an International Language*, 8(2), 88–101.
3. Ugorji, C.U.C., Iheanetu, O.U., Maduako, G., Mbakogu, C., Bickersteth, D. & Eyisi, J. (2024). *Automation of Preference Grammar*. Paper presented at the International Conference on Advancement of Engineering Innovation for Sustainable Development (ICAEISD 2024), Covenant University, Ota, 26–28 November 2024. Slides: the PG tenets, mechanism, components (input, parameters, output), the *think* operability table, and the proposed automation.

4. Jowitt, D. (2015). Nigerian Received Pronunciation. In T. Opeibi, J. Schmied, T. Omoniyi & K. Adedeji (eds.), *Essays on Language in Societal Transformation: A Festschrift in Honour of Segun Awonusi* (REAL Studies 9), 3–13. Göttingen: Cuvillier.
5. Prince, A. & Smolensky, P. (1993). *Optimality Theory: Constraint Interaction in Generative Grammar*. Technical Report RuCCS-TR-2, Rutgers University Center for Cognitive Science.
6. Jenkins, J. (2000). *The Phonology of English as an International Language*. Oxford: Oxford University Press.
7. Oyebola, F., Ho, S.Y.B. & Li, Z. (2019). A sociophonetic study on TH variation in educated Nigerian English. In *Proceedings of the 19th International Congress of Phonetic Sciences*, Melbourne, 2320–2323.
8. Bickersteth, O. (2025). *Development of a Pipeline for Continuous Speech Recognition and Transcription into IPA Representations*. BSc project, Department of Computer and Information Sciences, Covenant University, Ota. Supervisor: Dr O.U. Iheanetu.
9. Baevski, A., Zhou, H., Mohamed, A. & Auli, M. (2020). wav2vec 2.0: A framework for self-supervised learning of speech representations. *Advances in Neural Information Processing Systems* 33.
10. Xu, Q., Baevski, A. & Auli, M. (2022). Simple and effective zero-shot cross-lingual phoneme recognition. In *Proceedings of Interspeech 2022*. arXiv:2109.11680. (Model: `facebook/wav2vec2-xlsr-53-espeak-cv-ft`.)
11. Radford, A., Kim, J.W., Xu, T., Brockman, G., McLeavey, C. & Sutskever, I. (2023). Robust speech recognition via large-scale weak supervision. In *Proceedings of ICML 2023*. arXiv:2212.04356. (Model: `openai/whisper-small`.)
12. Desplanques, B., Thienpondt, J. & Demuynck, K. (2020). ECAPA-TDNN: Emphasized channel attention, propagation and aggregation in TDNN based speaker verification. In *Proceedings of Interspeech 2020*, 3830–3834.
13. Ravanelli, M. et al. (2021). SpeechBrain: A general-purpose speech toolkit. arXiv:2106.04624. (Model: `speechbrain/spkrec-ecapa-voxceleb`.)
14. Witt, S.M. & Young, S.J. (2000). Phone-level pronunciation scoring and assessment for interactive language learning. *Speech Communication*, 30(2–3), 95–108.
15. Cohen, J. (1960). A coefficient of agreement for nominal scales. *Educational and Psychological Measurement*, 20(1), 37–46.
16. Moran, S. & McCloy, D. (eds.) (2019). *PHOIBLE 2.0*. Jena: Max Planck Institute for the Science of Human History. https://phoible.org, doi:10.5281/zenodo.2626687.
17. Weinberger, S. (2015). *Speech Accent Archive*. George Mason University. http://accent.gmu.edu
18. Google (2019). *Crowdsourced high-quality Nigerian English speech data set* (SLR70). OpenSLR, https://openslr.org/70/ (CC BY-SA 4.0).
19. Hugging Face. *Spaces overview*. Hugging Face Hub documentation, https://huggingface.co/docs/hub/en/spaces-overview (read 20 October 2026). The paid-plan rule for new Docker Spaces was also returned by the Hub's own API on 20 October 2026: "hosting Gradio and Docker Spaces on free cpu-basic requires a PRO subscription".
20. Wolf, T. et al. (2020). Transformers: State-of-the-art natural language processing. In *Proceedings of EMNLP 2020: System Demonstrations*, 38–45.
21. Carnegie Mellon University. *The CMU Pronouncing Dictionary*. https://github.com/cmusphinx/cmudict
22. Llarena, J. *Britfone* (version 3.0.1), a British English pronunciation dictionary. https://github.com/JoseLlarena/Britfone
23. Reynolds, D.A. & Rose, R.C. (1995). Robust text-independent speaker identification using Gaussian mixture speaker models. *IEEE Transactions on Speech and Audio Processing*, 3(1), 72–83.

Project documents

[P1]	VoxPreference: Preference Grammar (PG) Tool. Product Requirement Document (PRD). Team document.
[P2]	A Framework for the Automation of Preference Grammar Analysis from Speech. Draft paper, PG automation team.
[P3]	User Flow for Preference Grammar System, and the earlier workflow diagram. Team diagrams (redrawn as Figure 1).
[P4]	Ten-stage pipeline plan for the automation of PG. Team presentation notes.

Evidence in the repository

Every test below runs from the repository with public data. The commands run in the Python environment from `backend/requirements.txt` (or `backend/requirements-intel-mac.txt` on an Intel Mac); the data scripts

(E1, E5) also need `pandas` and `remotezip`, and the audit's `--wer` option needs `jiwer` (in `backend/requirements-train.txt`): `pip install pandas remotezip jiwer`.

	Where	What it shows	How to rerun
E1	`backend/eval/fyp_audit.py`	Reads the training logs published with the fine-tuned model on Hugging Face (thebickersteth/wav2vec2-nigerian-english, runs/May14_12-26-44…); counts the label characters the original script produces; re-runs the original seeded split; with `--wer` scores the stock model on that split; with `--weights` compares every tensor with the stock model. Writes data/work/fyp_audit.json.	`cd backend && python -m eval.fyp_audit --wer --weights`
E2	`git history, commit e72090d`	The original backend/finetune_model.py (lowercased labels), backend/utils/split_shuffle_data.py (random 90/10 split, random_state=42) and backend/app.py (the /transcribe endpoint returns the transcription and IPA only).	`git show e72090d:backend/app.py` (and the other two paths)
E3	`backend/tests/test_tenets.py, backend/tests/test_ot.py`	Reproduce Ugorji's *think* and GOAT evaluations mark for mark, the *think* flip when [θ] is rare, expert overrides, the two readings of Frequency, and promoting, deleting and pooling tenets under strict domination. The tool prints the reason for each mark (e.g. the 78 word pairs for [t]).	`cd backend && python -m pytest tests/test_tenets.py tests/test_ot.py`
E4	`backend/eval/saa_gold.py, backend/eval/fit_calibration.py`	Scores the phone model against the phoneticians' transcriptions of the 22 Nigerian Speech Accent Archive speakers (753 tokens): accuracy, majority baseline and kappa per sound; keeps a calibration only where kappa rises on held-out speakers. The table is in backend/pg/README.md.	`python -m eval.fetch_saa; python -m eval.saa_gold; python -m eval.fit_calibration`
E5	`backend/eval/slr70.py and the pg command line`	Prepares the SLR70 inputs per sound and runs the analyses whose shares are reported in section 6.	`python -m eval.slr70`, then `python -m pg --unit TH-voiceless …` on the prepared lists
E6	`backend/eval/diarize_eval.py`	Builds conversations from SLR70 speakers whose identities are known and scores speaker separation: speakers found, share of words credited to the right speaker. Seed 0 was used for tuning, seed 1 is the held-out test.	`python -m eval.diarize_eval --conversations 8 --minutes 4 --seed 1`

E7	backend/pg/README.md, "Known limits"	The findings of the independent review: the model's lean towards dictionary forms, the exploratory status of acoustic sub-clusters and the contrast matrix, and why "same preferred form" is not an accuracy test.	read the file
E8	backend/tests/	The whole suite: alignment, phone scoring, tenets, OT evaluation, stability, outputs (CSV, VTT, JSON), speaker separation and the API.	cd backend && python -m pytest -q
E9	scripts/deploy_space.sh, Dockerfile	Builds and uploads the single image to the Hugging Face Space; the live checks in section 6 were made against https://thebickersteth-voxpreference.hf.space with POST /pg/analyze.	scripts/deploy_space.sh --push
E10	backend/tests/test_models.py	Runs six SLR70 clips through the real models: the variants and preferred form match the recorded reference, Whisper finds the target words, and a 3-speaker conversation is separated correctly. GitHub Actions runs it on every push before deploying (.github/workflows/ci-cd.yml).	cd backend && PG_MODEL_TESTS=1 python -m pytest -q tests/test_models.py